

A Full Negative Answer to the Four-Qubit Equality Question

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Source Question

The source is Andreas Bluhm, Anna Jencova, and Ion Nechita, *Incompatibility in general probabilistic theories, generalized spectrahedra, and tensor norms*, arXiv:2011.06497. In Section 12.2, the paper discusses the compatibility degree $\gamma(g; \text{QM}_2)$ for g dichotomic qubit measurements. It proves

$$\frac{1}{\sqrt{3}} \geq \gamma(g; \text{QM}_2) \geq \frac{1}{2} \quad (g \geq 4),$$

and records that

$$\lim_{g \rightarrow \infty} \gamma(g; \text{QM}_2) = \inf_{g \geq 1} \gamma(g; \text{QM}_2) = \frac{1}{\pi_1(\ell_2^3)} = \frac{1}{2}.$$

The paper then asks:

Question 12.1. Is the lower bound in Equation (47) optimal, that is,

$$\forall g \geq 4 \quad \gamma(g; \text{QM}_2) = \frac{1}{2} \iff \gamma(4; \text{QM}_2) = \frac{1}{2} ?$$

This recovers the results of [Bus86, BA07, BH08, PG11, BHSS13, BLPY16, BN18] (see Section 9.1 of [BN18] for the contributions of the respective works). In particular, the compatibility degree is $\gamma(g; \text{QM}_2) = 1/\sqrt{g}$ in this case.

For $g \geq 4$, we still have

$$\Gamma(g; \text{QM}_2) \supseteq \text{QC}_g,$$

which recovers the result of [BN18]. However, we find that the compatibility degree is bounded by

$$\forall g \geq 4, \quad 0.58 \approx \frac{1}{\sqrt{3}} \geq \gamma(g; \text{QM}_2) \geq \frac{1}{2}. \quad (47)$$

The upper bound follows from Equation (43) and the lower bound from the 1-summing norm of ℓ_2^3 . This improves over the lower bound $\max\{1/\sqrt{g}, 1/4\}$ obtained in [BN18] and recovers the best known upper bound imposed by the compatibility degree for $g = 3$ (see Proposition 6.13). Numerics using the SDP in [WPGF09] indicate that the SIC-POVM constructed by choosing vertices of a tetrahedron on the Bloch sphere has $\gamma(f) = 1/\sqrt{3}$, whereas we can find random examples with $\gamma(f) \leq 0.56 < 1/\sqrt{3}$ for $g = 4$. Thus, $\gamma(4; \text{QM}_2) < 1/\sqrt{3}$. However, the exact value of $\gamma(4; \text{QM}_2)$ remains presently open.

Question 12.1. *We know, from Proposition 11.5, that*

$$\lim_{g \rightarrow \infty} \gamma(g; \text{QM}_2) = \inf_{g \geq 1} \gamma(g; \text{QM}_2) = \frac{1}{\pi_1(\ell_2^3)} = \frac{1}{2}.$$

Is the lower bound in Equation (47) optimal, that is,

$$\forall g \geq 4 \quad \gamma(g; \text{QM}_2) = \frac{1}{2} \iff \gamma(4; \text{QM}_2) = \frac{1}{2} \quad ?$$

Beyond qubits, Proposition 11.7 gives

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Claimed Full Answer

The lower bound is not attained at $g = 4$:

$$\frac{1}{2} < \gamma(4; \text{QM}_2) \leq \frac{2}{\sqrt{13}} < \frac{1}{\sqrt{3}}.$$

Therefore the finite pointwise equality scenario in Question 12.1 is false. This packet gives a full negative answer to the stated equality question. It does not prove the exact value of $\gamma(4; \text{QM}_2)$; the explicit upper bound $2/\sqrt{13}$ is included because it appears to be the sharp value in numerical searches.

Definitions and Reduction Used

The source paper proves that qubits form the centrally symmetric GPT associated with the Bloch ball, i.e. with the Banach space ℓ_2^3 . Its Theorem 10.3 gives, for centrally symmetric GPTs,

$$\Gamma(g; V, V^+) = \left\{ s \in [0, 1]^g : \|s \cdot z\|_{\ell_1^g \otimes_\pi \bar{V}} \leq 1 \text{ for all } \|z\|_{\ell_1^g \otimes_\pi \bar{V}} \leq 1 \right\}.$$

For $X = \ell_2^3$, $z = \sum_{i=1}^4 e_i \otimes z_i \in \ell_1^4 \otimes X$, one has

$$\|z\|_{\ell_1^4 \otimes_\pi X} = \sum_{i=1}^4 \|z_i\|_2$$

and

$$\|z\|_{\ell_1^4 \otimes_\varepsilon X} = \max_{\varepsilon_i = \pm 1} \left\| \sum_{i=1}^4 \varepsilon_i z_i \right\|_2.$$

Consequently

$$\gamma(4; \text{QM}_2) = \inf_{(z_i) \neq 0} \frac{\max_{\varepsilon_i = \pm 1} \left\| \sum_{i=1}^4 \varepsilon_i z_i \right\|_2}{\sum_{i=1}^4 \|z_i\|_2},$$

where $z_1, \dots, z_4 \in \mathbb{R}^3$.

Proof Intuition

The source lower bound $1/2$ comes from a dimension-free 1-summing estimate. Equality at four vectors would be extremely rigid. After normalizing the sum of the four lengths to one, the average squared length of a random signed sum is $\sum_i \|z_i\|^2$, which is at least $1/4$ by Cauchy's inequality. If the maximum signed sum had norm only $1/2$, every inequality in this averaging argument would have to be equality. That forces all four lengths to be equal and all signed sums to have the same squared norm. The signed-square expansion then says that every pairwise inner product is zero, which would give four nonzero orthogonal vectors in $\mathbb{R}^3$.

Theorem and Proof

Theorem 1.

$$\frac{1}{2} < \gamma(4; \text{QM}_2) \leq \frac{2}{\sqrt{13}} < \frac{1}{\sqrt{3}}.$$

In particular, the equality $\gamma(4; \text{QM}_2) = 1/2$ asked about in Question 12.1 is false.

Proof. Using the reduction above, define

$$R(z_1, z_2, z_3, z_4) = \frac{\max_{\varepsilon_i = \pm 1} \left\| \sum_{i=1}^4 \varepsilon_i z_i \right\|_2}{\sum_{i=1}^4 \|z_i\|_2}$$

for nonzero four-tuples in $(\mathbb{R}^3)^4$. The source paper already gives $\gamma(4; \text{QM}_2) \geq 1/2$. Suppose, for contradiction, that equality holds.

Normalize $\sum_i \|z_i\|_2 = 1$. Since the normalized set of four-tuples is compact and the function

$$(z_i)_{i=1}^4 \mapsto \max_{\varepsilon_i = \pm 1} \left\| \sum_{i=1}^4 \varepsilon_i z_i \right\|_2$$

is continuous, equality at the infimum is attained by some tuple (z_1, z_2, z_3, z_4) . Thus

$$\max_{\varepsilon_i = \pm 1} \left\| \sum_{i=1}^4 \varepsilon_i z_i \right\|_2 = \frac{1}{2}.$$

Let $r_i = \|z_i\|_2$. For independent Rademacher signs ε_i ,

$$\mathbb{E} \left\| \sum_{i=1}^4 \varepsilon_i z_i \right\|_2^2 = \sum_{i=1}^4 \|z_i\|_2^2 = \sum_{i=1}^4 r_i^2.$$

Because every signed sum has norm at most $1/2$, this gives

$$\sum_{i=1}^4 r_i^2 \leq \frac{1}{4}.$$

On the other hand, $\sum_i r_i = 1$, so Cauchy's inequality gives

$$\sum_{i=1}^4 r_i^2 \geq \frac{1}{4} \left(\sum_{i=1}^4 r_i \right)^2 = \frac{1}{4}.$$

Hence equality holds throughout. In particular

$$r_1 = r_2 = r_3 = r_4 = \frac{1}{4},$$

so all four vectors are nonzero, and since the average of the squared signed norms is $1/4$ while each squared signed norm is at most $1/4$, every signed sum satisfies

$$\left\| \sum_{i=1}^4 \varepsilon_i z_i \right\|_2^2 = \frac{1}{4}.$$

Expanding the square gives, for every sign vector $\varepsilon \in \{\pm 1\}^4$,

$$\frac{1}{4} = \sum_{i=1}^4 \|z_i\|_2^2 + 2 \sum_{1 \leq i < j \leq 4} \varepsilon_i \varepsilon_j \langle z_i, z_j \rangle.$$

Since $\sum_i \|z_i\|_2^2 = 1/4$, this reduces to

$$\sum_{1 \leq i < j \leq 4} \varepsilon_i \varepsilon_j \langle z_i, z_j \rangle = 0 \quad (\varepsilon \in \{\pm 1\}^4).$$

The functions $\varepsilon \mapsto \varepsilon_i \varepsilon_j$ are linearly independent characters on the sign cube. Therefore every coefficient must vanish:

$$\langle z_i, z_j \rangle = 0 \quad (i \neq j).$$

This is impossible, because $\mathbb{R}^3$ cannot contain four nonzero pairwise orthogonal vectors. The contradiction proves that $\gamma(4; \text{QM}_2) > 1/2$.

It remains to prove the advertised explicit upper bound. Let

$$a = \frac{3}{2\sqrt{13}}, \quad b = \frac{2}{\sqrt{13}},$$

and take

$$\begin{aligned} p_1 &= a(1, 0, 0), \\ p_2 &= a(-1/2, \sqrt{3}/2, 0), \\ p_3 &= a(-1/2, -\sqrt{3}/2, 0), \\ p_4 &= b(0, 0, 1). \end{aligned}$$

Then $p_1 + p_2 + p_3 = 0$, the first three vectors have common length a , and p_4 is orthogonal to their plane. For any signs on p_1, p_2, p_3 , their signed sum is either zero, if all three signs are equal, or has

length $2a = 3/\sqrt{13}$, if the signs are not all equal. Hence every signed sum of p_1, p_2, p_3, p_4 has norm at most

$$\max \left\{ b, \sqrt{(2a)^2 + b^2} \right\} = \max \left\{ \frac{2}{\sqrt{13}}, \sqrt{\frac{9}{13} + \frac{4}{13}} \right\} = 1.$$

The total generator length is

$$\sum_{i=1}^4 \|p_i\|_2 = 3a + b = \frac{9}{2\sqrt{13}} + \frac{2}{\sqrt{13}} = \frac{\sqrt{13}}{2}.$$

Using the sign-sum formula for $\gamma(4; \text{QM}_2)$, this four-tuple gives

$$\gamma(4; \text{QM}_2) \leq \frac{1}{\sqrt{13}/2} = \frac{2}{\sqrt{13}}.$$

Finally $2/\sqrt{13} < 1/\sqrt{3}$, since $12 < 13$. □

Consequences, Exact-Value Candidate, and Limitations

Question 12.1's proposed finite equality scenario fails at $g = 4$. The explicit construction also improves the source paper's reported numerical examples near 0.56 to the exact closed-form bound $2/\sqrt{13} \approx 0.5547001962$.

The exact value is still not proved in this packet. A direct numerical optimization of the equivalent zonotope problem repeatedly returns the same value $2/\sqrt{13}$, with extremal configuration equivalent to the three coplanar 120° -spaced generators plus one orthogonal generator above. Thus the natural remaining conjecture is

$$\gamma(4; \text{QM}_2) = \frac{2}{\sqrt{13}},$$

equivalently: every four-generator centrally symmetric zonotope contained in the Euclidean unit ball of $\mathbb{R}^3$ has total generator length at most $\sqrt{13}/2$.

The source paper's asymptotic statement remains compatible with this result: $\gamma(g; \text{QM}_2)$ may still decrease to $1/2$ as $g \rightarrow \infty$.

Novelty and Duplicate Checks

Before promotion I searched the cheap run indexes for arXiv 2011.06497, the title, **generalized spectrahedra**, **tensor norms**, **QM_2**, **gamma(4)**, **compatibility degree**, and nearby qubit compatibility phrases. No exact packet or attempt hit was present.

I also ran bounded web and literature searches for exact and near-exact phrases including

- "gamma(4; QM_2)" "1/2",
- "compatibility degree" "QM_2" "1/2" "gamma(4",
- "four qubit" "compatibility degree" "1/2",
- "joint measurability" "four" "qubit" "1/2",
- "four generator zonotope" "Euclidean ball",

- "vector balancing" "four vectors" "R3" "signs",
- "projective injective tensor norm" "ell1-4" "ell2-3".

The search found standard nearby themes—three-observable qubit compatibility criteria, vector balancing, and modern zonotope containment—but did not surface a later explicit resolution of Question 12.1 or an exact computation of $\gamma(4; \text{QM}_2)$. The result should nevertheless be reviewed against the quantum-measurement and convex-geometry literature.

References

References

- [1] A. Bluhm, A. Jencova, and I. Nechita, *Incompatibility in general probabilistic theories, generalized spectrahedra, and tensor norms*, arXiv:2011.06497.
- [2] S. Yu and C. H. Oh, *Quantum contextuality and joint measurement of three observables of a qubit*, arXiv:1312.6470.
- [3] F. Eisenbrand, T. Rothvoss, L. Russo, and M. Skorupinski, *Nearly-Tight Bounds for Zonotope Containment and Beyond*, arXiv:2605.04183.

Human Review Recommendation

Review should focus on the reduction of $\gamma(4; \text{QM}_2)$ to the four-vector sign-sum formula. Once that reduction is accepted from the source paper's centrally symmetric GPT theorem, the strictness proof and the explicit $2/\sqrt{13}$ construction are elementary finite-dimensional arguments. The exact-value conjecture is not claimed as proved here.